① Known: radius R

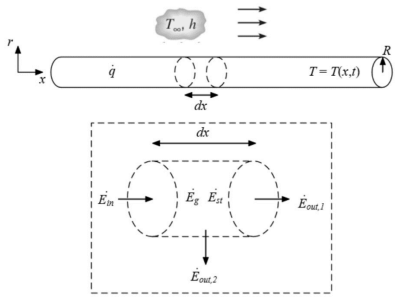
T_∞, h

Find a) energy balance for diff eq.

b) governing diff eq.

c) boundary & initial condition

Schematic:



Assumptions:
→ 1-D conduction in x
→ radial temperature uniform
→ constant properties, no radiation
→ convection only from lateral surface

Analysis:

a) $q_x = -kA\frac{dT}{dx}$

@ $x \to \dot{E}_{in} = -kA\frac{dT}{dx}$

@ $x+dx$ $\quad \dot{E}_{out} = \left(-kA\frac{dT}{dx} + \frac{d}{dx}\left(-kA\frac{dT}{dx}\right)dx\right)$

$\dot{E}_g = \dot{q}(Adx)$

Convection losses → $A_s = Pdx = (2\pi R)dx$

$\dot{E}_{conv} = hPdx(T-T_\infty)$

$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g - \dot{E}_{conv} = \dot{E}_{acc}$

$$kA\frac{d^2T}{dx^2}dx + \dot{q}Adx - hPdx(T-T_\infty) = \rho c_p A dx\frac{dT}{dt}$$

$$k\frac{d^2T}{dx^2} + \dot{q} - \frac{hP}{A}(T-T_\infty) = \rho c_p\frac{dT}{dt}$$

b) $\frac{P}{A} = \frac{2\pi R}{\pi R^2} = \frac{2}{R}$

$$\rho c_p\frac{\partial T}{\partial x} = k\frac{\partial^2 T}{\partial x^2} + \dot{q} - \frac{2h}{R}(T-T_\infty)$$

c) @ $x=0 \to T(0,t) = T_1$

@ $x=L$

$-k\frac{dT}{dx} = h(T-T_\infty)$

$$-k\frac{dT}{dx}\Big|_{x=L} = h(T(L,t)-T_\infty)$$

initial condition

$T(x,0) = T_i$